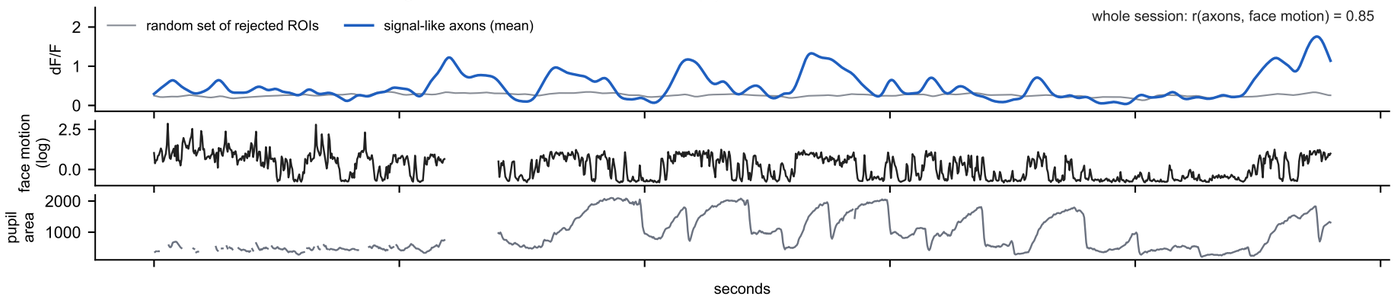
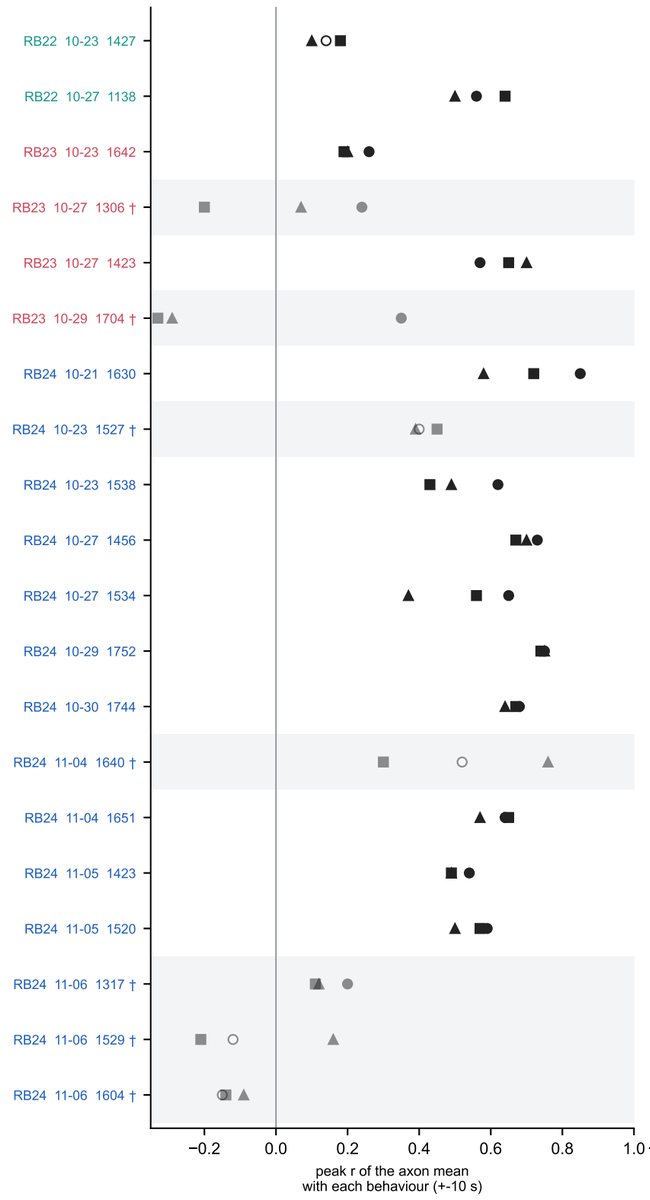
RB22 - RB24 LC axons in the superior colliculus: what the data support

20 sessions, 3 mice (RB24 14, RB23 4, RB22 2 sessions with accepted axon ROIs); 1-plane and 5-plane imaging; exploratory, one pass. Signal-like = ROIs whose trace passes the V3 axon criterion; "movement removed" = a behaviour-only model (face + whisking + ball) subtracted.

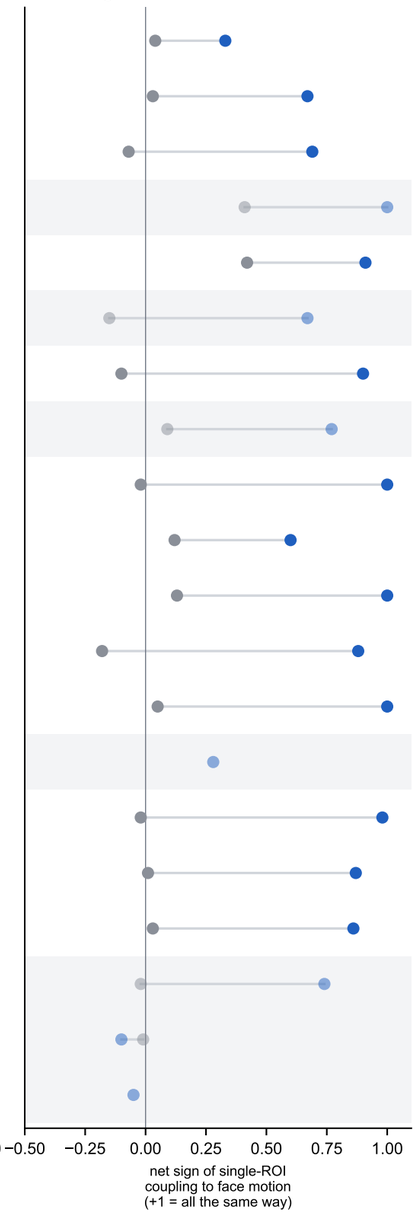
A The shared arousal / movement signal (RB24 10-21 1630, 4 min excerpt)



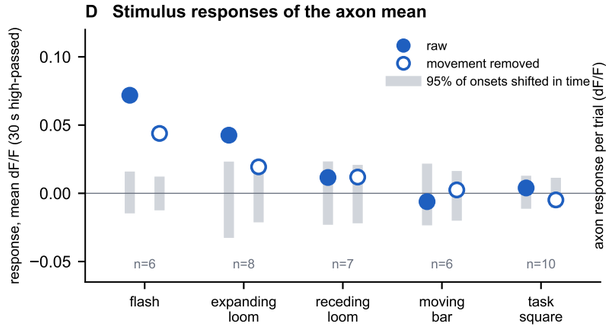
B Coupling to behaviour



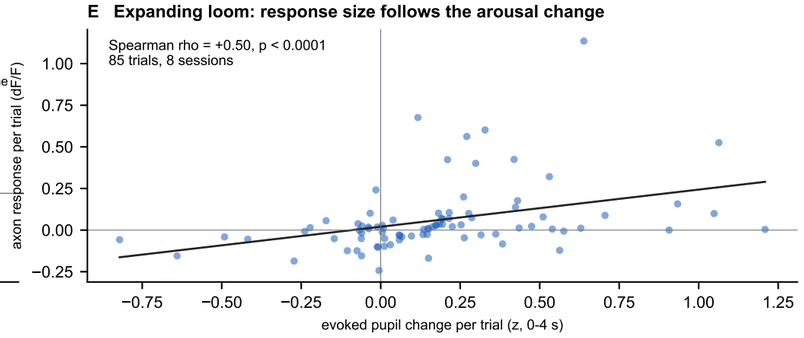
C ROI sign



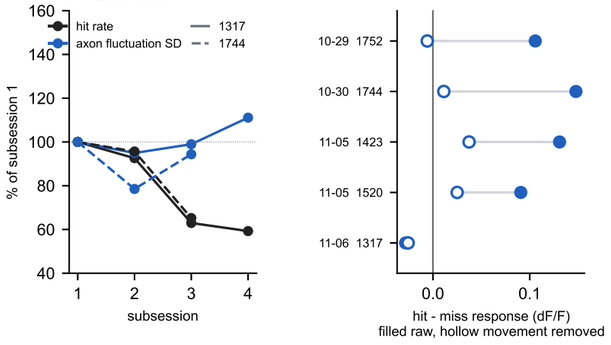
D Stimulus responses of the axon mean



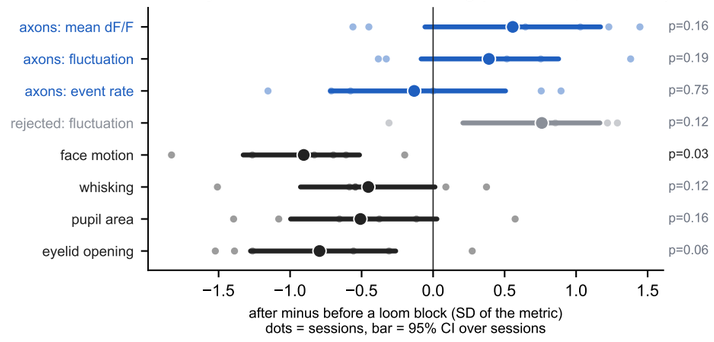
E Expanding loom: response size follows the arousal change



F Lick task



G Grey screen before vs after looming (6 sessions, 21 blocks)



H How much weight each result can carry

STRONG — **Shared arousal / movement signal in the axons.**
11 of 20 sessions, 3 mice: signal-like ROIs share ONE signal (PC1 45-80% of variance) that follows face motion, whisking and ball at lag 0 (pupil +1.2 s, slow kinetics); the ROIs the experimenter rejected do not (sign test, panel C).

MODERATE — **Axons respond to flashes beyond movement.**
+0.07 dF/F (6 sessions, p 0.001), +0.044 after removing movement (p 0.001). Flashes also constrict the pupil (-1.6 SD), so a residual arousal / light-reflex contribution is not excluded.

WEAK — **Expanding-loom response.**
+0.043 dF/F (8 sessions, p 0.005), +0.019 after removing movement (p 0.06); per trial it scales with the evoked pupil change (rho +0.50). Receding disc, moving bar, task square: nothing.

NOT SEEN — **Anything specific to looming, reward, task engagement, or the grey after looming.**
Hit - miss differences vanish once movement is removed; in 1317 the hit rate falls to 59% of its start while axon fluctuations stay flat; 21 loom blocks show no axon change (behaviour calms, axons do not follow).

OPEN — **The three 11-06 sessions (1529, 1317, 1604) show no coupling.**
Behaviour barely varies there (face-motion SD 0.17-0.24 vs 0.4-0.8 earlier) and the shared signal is weaker (PC1 0.17-0.58): rig / FOV / expression change, or just a quiet, habituated animal?

- B: face motion (filled = p < 0.01 vs shifted data)
- B: whisking
- B: pupil
- C: signal-like ROIs, net sign of the single-ROI coupling to face motion
- C: rejected ROIs (random mix of signs = no shared signal)

11 of 20 sessions are clearly coupled (r >= 0.4, p < 0.01 against circularly shifted data); the 2 short sessions (1527, 1640) go the same way. † shaded rows: low behavioural range (11-06), inconsistent face / whisking / pupil tracking (RB23 1306, 1704) or very short sessions.

I What limits every conclusion above (fix before the next experiments)

No reference channel

Only GCaMP is imaged: axial (z) motion and neuropil cannot be separated from signal. The 5-plane z-registration failed (planes almost identical at 3 um steps).

Little behavioural variance

Habituated animals move little (11-06); the state signal needs arousal transitions to be measured. The pupil is confounded by screen luminance (flashes, breaks).

Weak logs on the rig

Ball log truncated in open-loop blocks, 2nd ball sensor dead, 3.5% projector sync loss, lick line not on an aux channel, pump TTL not logged, rig-code reward rule unconfirmed.

Design

No no-loom control block, 5-20 trials per class, fixed block order, grey screens sometimes declared but not shown, ROI curation not blind to the traces, axon identity / histology not established.